

Charotar University of Science and Technology
Education Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY)
SEMESTER I EXAMINATION (BACKLOG)
BC701: GENERAL MICROBIOLOGY AND BIOCHEMISTRY

Date: 03-05- 2010

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A

Total marks: 20

Q1. A. Choose the correct option and put $\sqrt$ mark in front of it:

1. The following is the major characteristic of bacteria :
(a) prokaryotes (b) have peptidoglycan cell wall
(c) grow by binary fission (d) all of the above
2. Use of microorganisms to selectively recover metals from low grade ore is known as
(a) biodegradation (b) biosynthesis (c) bioleaching (d) biorecovery
3. The following is **NOT** the distinguishing feature of a fungal cell
(a) DNA associated with histones (b) 18S rRNA
(c) a single circular chromosome (d) all of the above
4. The arrangement of organisms into groups whose members share many characteristics and reflect many of their biological nature is termed as
(a) Taxonomy (b) phylogenetic classification
(c) Natural classification (d) all of the above
5. The following technique is used for the determination of microbial growth
(a) cell forming units (b) microbial dry weight (c) spectrophotometry (d) all of the above
6. Mean doubling time of an organism growing as a batch culture is equivalent to
(a) doubling time (b) Generation time
(c) time required for DNA replication (d) all of the above
7. Exopolysaccharides are produced by some microorganisms which help in the formation of
(a) biofilms (b) colonies (c) lichen (d) all of the above
8. The turbidostat operates best at the following dilution rates
(a) high (b) low (c) intermediate (d) all of the above
9. The Glycosaminoglycan that serves as an anticoagulant is known as
(a) Hyaluronic acid (b) Chondroitin sulfate (c) Dermatan sulfate (d) none of the above

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10. The nitrogen base present in lecithin is
(a) Choline (b) Ethanol amine (c) Inositol (d) serine
11. Most abundant animal sterol is
(a) Vit-D (b) Ergosterol (c) stigmasterol (d) none of the above
12. The following is **NOT** true for the structure of water
(a) cannot form intermolecular hydrogen bonds (b) forms flickering clusters
(c) has a distorted tetrahedral atomic arrangement (d) none of the above
13. The following Polysaccharide is **NOT** composed of fructose.
(a) Starch (b) Glycogen (c) Dextrin (d) all of the above
14. The following is a non-protein amino acid
(a) Ornithine (b) Homocysteine (c) Histidine (d) Lysine
15. The following best describes the mode of action of enzymes?
(a) decreasing the activation energy of the reaction (b) altering the reaction equilibrium
(c) increasing the activation energy of the reaction (d) none of the above
16. The following reaction will tend to occur spontaneously if ΔG is negative,
(a) Forward (b) Reverse (c) Both sides (d) Equilibrium
17. One non-spontaneous reaction is driven forward by the following coupled reaction
(a) non spontaneous reaction (b) dehydrogenase reaction
(c) hydrolytic reaction (d) none of the above
18. Carbon monoxide inhibits electron transport chain by
(a) Inhibiting F_0 (b) inhibiting cytochrome oxidase
(c) preventing transfer from Fe-S to Ubiquinone (d) all of the above
19. The following **is** the strongest interaction occurring in the biomolecules
(a) London-dispersion forces (b) Van der Waal's forces (c) covalent bonds (d) hydrogen bonds
20. The following **is not** an epimer of D- glucose:
(a) D- mannose (b) D-fructose (c) D- arabinose (d) all of the above

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2. Write answers for Section I and Section II of **Part B** in separate answer sheets.

Part B	Total Marks: 50
Section I	
Q.1 (A) Write a note on ANY TWO of the following: (i) The cycling of nitrogen by microbes (ii) Bioleaching (iii) Multilocus sequence typing	(96)
Q.1 (B) Differentiate between ANY TWO the following: (i) Symbiotic and non-symbiotic nitrogen fixation (ii) FISH and FAME (iii) Prokaryotes and eukaryotes	(04)
Q.2 (A) Differentiate between ANY TWO the following: (i) Heterotrophs and autotrophs (ii) Batch and continuous growth (iii) Enrichment and enriched media	(04)
Q.2 (B) Define the following: (ANY SIX) (i) Chemostat (ii) Turbidostat (iii) Defined media (iv) Selective Media (v) Differential media (vi) Growth (vii) Generation time (viii) Exponential growth phase	(06)
Section: II	
Q.3 (A) Attempt ANY ONE of the following: (i) Define and give examples of buffers. How buffers resist change in pH. (ii) Discuss various types of weak and covalent bonds present in three dimensional structure of proteins	(06)

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- Q.3 (B)** Differentiate between: Covalent bond and weak bonds (04)
- Q.4 (A)** Differentiate between **ANY TWO** of the following: (06)
- (i) Essential amino acids and non-essential amino acids
 - (ii) Reducing sugar and non-reducing sugar
 - (iii) Essential fatty acids and non-essential fatty acids
- Q.4 (B)** Define/explain **ANY TWO** of the following terms: (04)
- (i) Acid value of fat
 - (ii) Iodine number of fat
 - (iii) Hydropathy index of protein
- Q.5 (A)** Distinguish between **ANY TWO** of the following (06)
- (i) Oxidative phosphorylation and Substrate level phosphorylation
 - (ii) High energy compounds and Low energy compounds
 - (iii) Free energy change and standard free energy change
- Q.5 (B)** Which are the different types of electron transfers involved in biological systems? (04)